

Econ 527 HW3

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1 Speeding up VFI

This problem is the same form last problem set i.e. We seek to solve a simple representative agent neoclassical growth model. An agent solves

$$\begin{aligned} \max_{\{c_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t) \\ \text{s.t. } k_{t+1} = Ak_t^\alpha + (1 - \delta)k_t - c_t \end{aligned} \tag{1}$$

where $u(c) = \log(c)$, $k_t > 0$, and k_0 is given. The problem 1 can be written as

$$V(k) = \max_{k' \geq 0} \log(Ak^\alpha + (1 - \delta)k - k') + \beta V(k') \tag{2}$$

We set the following parameters: $\beta = 0.96$, $A = 1$, $\delta = 1$, and $\alpha = 0.35$.

1.1 Howard Policy Iteration

To speed up Value Function Iteration, we implement the Howard Policy Iteration algorithm.

Algorithm 1 Howard Policy Iteration Algorithm

Require: Grid $\{a_i\}_{i=1}^N$, income states $\{z_j\}$, parameters $\beta, r, a_{\min}$, tolerance ϵ , iterations M

1: Guess $V^0(a_i, z_j)$, set $n = 0$

2: **repeat**

3: Construct interpolants from $\{a_i, V^n(a_i, z_j)\}$ and perform one Bellman iteration:

$$V^{n+1}(a_i, z_j) = \max_{a' \geq a_{\min}} \{u((1+r)a_i + \exp(z_j) - a') + \beta \mathbb{E}[V^n(a', z') \mid z_j]\}$$

4: **if** converged **then**

5: **stop**

6: **else**

7: Save optimizer $a'(a_i, z_j)$

8: Set $\mathcal{V}^0(a_i, z_j) = V^{n+1}(a_i, z_j)$, set $m = 0$

9: **repeat**

10: Construct interpolants and perform one Bellman iteration:

$$V^{n+1}(a_i, z_j) = \max_{a' \geq a_{\min}} \{u((1+r)a_i + \exp(z_j) - a') + \beta \mathbb{E}[V^n(a', z') \mid z_j]\}$$

11: $m \leftarrow m + 1$

12: **until** $m = M - 1$

13: Set $V^{n+1}(a_i, z_j) = \mathcal{V}^M(a_i, z_j)$

14: **end if**

15: $n \leftarrow n + 1$

16: **until** $\|V^{n+1} - V^n\|_{\infty} < \epsilon(1 - \beta)$

The results are

M	Time (s)	Bellman iterations
5	9.404590667	34
10	7.969550042	23
20	7.152319792	15

Table 1: Results from Howard Policy Improvement

1.2 Endogeneous Grid Method

For the Endogeneous Grid Method, it took **2433** iterations and took **1.43** seconds.

2 The Stationary Distribution of State Variables

2.1 Net Demand Using Interpolation

The transition matrix was calculated via cubic spline interpolation. Using the current wealth grid from Endogeneous grid pint method, cubic spline interpolants were calculated to fit a finer wealth grid (defined as 1,000 evenly spaced gridpoints over the range $[a_{min}, a_{max}]$) on the policy grid to approximate the policy function $g(a_i, z_j)$.

Given the policy function $g(a_i, z_j)$ and transition matrix of z , the pair (a, z) follows a Markov process with the transition matrix Υ with dimensions $ij \times ij$. The stationary distribution is then calculated using the eigenvalue method such that

$$\Upsilon' = \Upsilon' \phi(a_i, z_j) \quad (3)$$

and $\sum_i v_i = 1$, where $\phi(a_i, z_j)$ is the stationary distribution with dimensions $i \times j$. We can write (3) as a system of inear equations

$$\begin{pmatrix} v_{11} & v_{12} & \cdots & v_{1i} \\ v_{21} & v_{22} & \cdots & v_{2i} \\ \vdots & \ddots & & \vdots \\ v_{j1} & v_{j2} & \cdots & v_{ji} \end{pmatrix} = \begin{pmatrix} v_{11} & v_{12} & \cdots & v_{1i} \\ v_{21} & v_{22} & \cdots & v_{2i} \\ \vdots & \ddots & & \vdots \\ v_{j1} & v_{j2} & \cdots & v_{ji} \end{pmatrix} \phi(a_i, z_j)$$

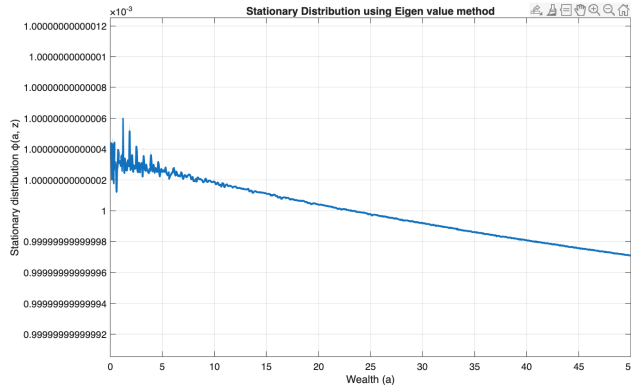


Figure 1: Stationary distribution of state variables vs. wealth

According to 1, the stationaty distribution equals 1 only once i.e.

$$P$$

2.2 Net Demand Using Simulation